Exact variance on arbitrary finite step supports

Lean proves that every finite abelian group, every distinct finite increment table and every nonnegative real label law summing to 1 has actual in-degree mean 1 and variance $1 - \sum_i p_i^2$ at every target, under the genuine independent-vertex-label PMF. The theorem covers all finite group sizes and all such probability laws. [pcc:unverified]

The whole expanded walk project gate passes **70 declarations**, zero failures and zero trusted-native exceptions. Only propext, Classical.choice and Quot.sound are accepted. Lean 4.35.0 and Mathlib revision prefix `56a00585` are pinned. No custom axiom or proof hole. [pcc:unverified]

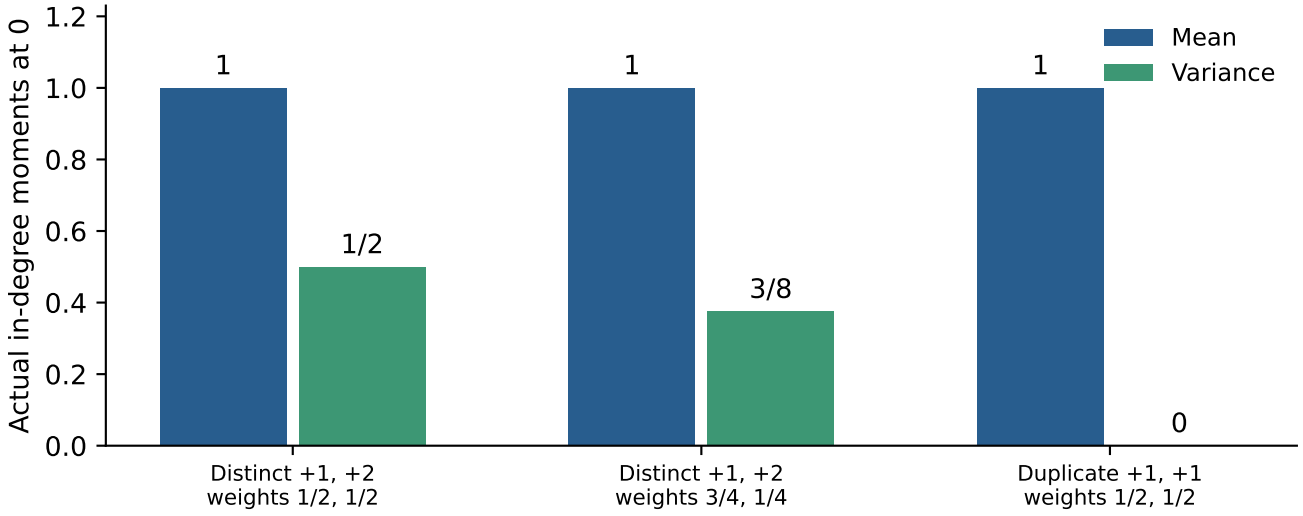


Figure 1: Exact kernel controls on Z/3. The duplicate-step control deliberately violates distinctness. No sampling, uncertainty interval, runtime or collision-time inference.

Lean directly disproves replacing distinctness by the weaker requirement that every increment is nonzero. Two labels both selecting +1 on Z/3 produce a translation for every hash: the actual in-degree is always 1, so variance is 0. For equal label probabilities, the unmerged-label expression $1 - \sum_i p_i^2$ is 1/2. The group, probability law and nonzero increments are preserved; the failed hypothesis is explicit. [pcc:unverified]

This is a negative control for an omitted assumption. It preserves the qualified distinct-support theorem and does not refute the collision-time heuristic. [pcc:unverified]

Actual graph, product probability law and proof

For $h : G \to I$, the graph is $x \mapsto x + g_{h(x)}$ and the PMF mass of h is $\prod_{x \in G} p_{h(x)}$. A finite product-of-sums identity proves normalization; nonnegativity gives an actual PMF. Coordinate-product identities establish exact independence at different vertices. [pcc:unverified]

The incoming-edge cardinality at target z is proved equal to the sum of indicators $h(z - g_i) = i$. Its mean is 1. Distinct increments give distinct predecessors for different labels, allowing the second moment to split into equal-label and independent-label terms. Finite ring algebra yields $2 - \sum_i p_i^2$ for the second moment and $1 - \sum_i p_i^2$ for variance. No graph table, empirical fit or collision-time law is assumed. [pcc:unverified]

The main binding explicitly quantifies types, group and finite structures, label weights, normalization, step injectivity and target. It combines the universal moment theorem with the direct nonzero-only counterexample: `general_variance_audit_certificate` in WalkWeightedGeneralAudit. The source and proof map are linked from `../../WEIGHTED_GENERAL.md`. [pcc:unverified]

Source commit `6f963444` precedes theorem packet `PCC-20261110-4e62d0`. Its full local check passes at 01:29:41 UTC on producer host `762f5fc120c9`; signed standing remains unverified. Exact statement/source hashes, the whole gate, failed sources and diagnostic gzip archives are in `../../general-variance-sources.json` and evidence/. [pcc:unverified]

The collision-time factor $1/\sqrt{1 - \sum_i p_i^2}$, limiting constants, Hildebrand's random-support averaging and practical hash-family behavior remain separate. No new native experiment, timing or paid compute was run. Earlier issued proof/native/protocol sources remain frozen. [pcc:unverified]

Primary Galbraith chapter 14, version 2, section 14.2.5, pages 296–297, was re-read at the recorded scope in weighted-sources.json, including its ideal random-function assumption and separate conjectural collision estimates. No novelty claim is made. [pcc:unverified]